1 Introduction

In the definition of $\int_a^b f(x)dx$, we assume that

- f is defined on an interval $[a,b]$ of finite length, and
- f is bounded on $[a,b]$.

However, there are situations in which we would like to relax these assumptions. Here are some examples.

Example 1.1 (Gravitational Escape)**.** Suppose we launch an object of mass m from the surface of the Earth to an altitude H. The energy we must impart to the object in order for it to reach this height is given by

$$\int_R^{R+H} \frac{GMm}{x^2}dx,$$

where R is the radius of the Earth, M the mass of the Earth, and G Newton's gravitational constant. How much energy does it take for the object to escape Earth's gravity so that it continues to float away forever and never return? In this case, H can be arbitrarily large, so to calculate the energy we need to make sense of the integral

$$\int_R^{\infty} \frac{GMm}{x^2}dx.$$

Example 1.2 (Arc length)**.** Suppose you forgot the formula for the circumference of a circle, but you remember the formula for arc length:

$$\int_a^b \sqrt{1+(f'(x))^2}dx.$$

Realizing the upper semicircle of the unit circle centered at the origin as the graph of the function $y = f(x) = \sqrt{1-x^2}$, differentiating gives

$$(f'(x))^2 = \left(\frac{-x}{\sqrt{1-x^2}}\right)^2 = \frac{x^2}{1-x^2},$$

so the circumference is given by twice the integral

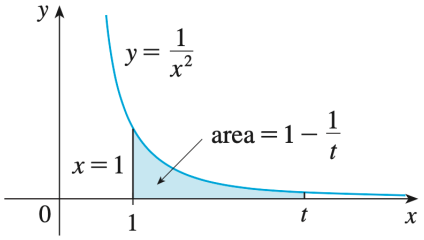
$$\begin{aligned}\int_{-1}^{1} \sqrt{1+\frac{x^2}{1-x^2}}dx &= \int_{-1}^{1} \sqrt{\frac{1-x^2}{1-x^2}+\frac{x^2}{1-x^2}}dx \\ &= \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}dx.\end{aligned}$$

Unfortunately, the integrand is not bounded on $[-1,1]$ as it has vertical asymptotes at the endpoints, so the integral does not exist, even though we would like it to somehow be equal to π.

2 Improper Integrals of Type 1: Infinite Interval

Ignoring the constants in the first example and setting $R = 1$, the integral we would like to evaluate is $\int_1^\infty \frac{1}{x} dx$. This represents the area of the region S under the graph of the function $y = \frac{1}{x^2}$, above the x-axis, and to the right of the line $x = 1$. One might think that, since S is infinite in extent, its area must be infinite, but let's take a closer look. If we fix any $t > 1$, then the area of the part of S that lies to the left of the line $x = t$ (shown in the figure below) is

$$A(t) = \int_1^t \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^t = 1 - \frac{1}{t}.$$

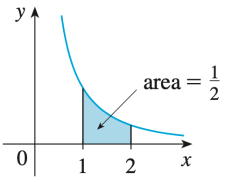


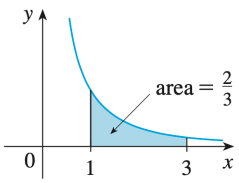
Notice that $A(t) < 1$ no matter low large t is chosen. We also observe that

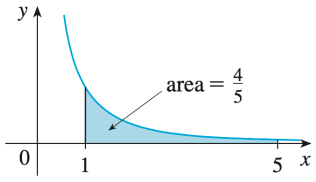
$$\lim_{t \to \infty} A(t) = \lim_{t \to \infty} \left(1 - \frac{1}{t} \right) = 1.$$

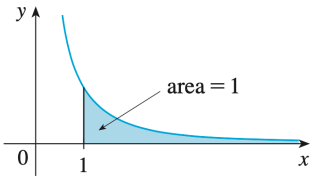
The area of the shaded region approaches 1 as $t \to \infty$, so we say the area of the infinite region S is equal to 1 and we write

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{t \to \infty} \int_1^t \frac{1}{x^2} dx = 1.$$









Motivated by this example, we make the following definition:

Definition 2.1 (Improper Integrals of Type 1)**.**

(a) If $\int_a^t f(x)dx$ exists for every $t \geq a$, then we define

$$\int_a^\infty f(x)dx := \lim_{t\to\infty}\int_a^t f(x)dx,$$

provided this limit exists (as a finite number).

(b) If $\int_t^b f(x)dx$ exists for every $t \leq b$, then we define

$$\int_{-\infty}^b f(x)dx := \lim_{t\to-\infty}\int_t^b f(x)dx,$$

provided this limit exists (as a finite number).

The *improper integrals* $\int_a^\infty f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called *convergent* if the corresponding limit exists and *divergent* if the limit does not exist.

(c) If both $\int_a^\infty f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent, then we define

$$\begin{aligned}\int_{-\infty}^\infty f(x)dx &= \int_{-\infty}^a f(x)dx + \int_a^\infty f(x)dx \\ &= \lim_{s\to-\infty}\int_s^a f(x)dx + \lim_{t\to\infty}\int_a^t f(x)dx.\end{aligned}$$

Remark 2.2. Two comments about part (c):

- $\int_{-\infty}^\infty f(x)dx$ is independent of the choice of a, so any real number a can be used.
- $\int_{-\infty}^\infty f(x)dx$ is defined to be $\lim_{s\to-\infty}\int_s^a f(x)dx + \lim_{t\to\infty}\int_a^t f(x)dx$, **not** $\lim_{t\to\infty}\int_{-t}^t f(x)dx$. That is, the two limits are taken independently.

1. Determine whether the integral $\int_1^\infty \frac{dx}{x}$ is convergent or divergent.

We have just seen that

$$\int_1^\infty \frac{dx}{x^2} \text{ converges} \qquad \int_1^\infty \frac{dx}{x} \text{ diverges.}$$

Geometrically, this says that although the curves $y = \frac{1}{x^2}$ and $y = \frac{1}{x}$ look very similar for $x > 0$, the region under $y = \frac{1}{x^2}$ to the right of $x = 1$ has finite area whereas the corresponding region under $y = \frac{1}{x}$ has infinite area.

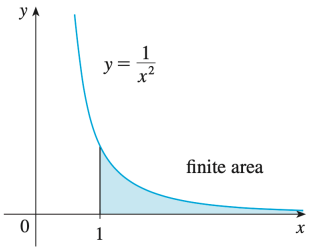


FIGURE 4 $\int_1^\infty (1/x^2)\,dx$ converges

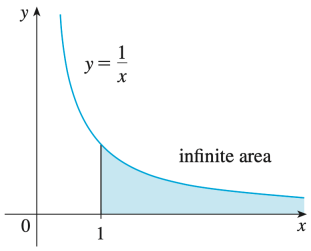


FIGURE 5 $\int_1^\infty (1/x)\,dx$ diverges

Both $\frac{1}{x^2}$ and $\frac{1}{x}$ approach 0 as $x \to \infty$, but $\frac{1}{x^2}$ approaches 0 faster than $\frac{1}{x}$. The values of $\frac{1}{x}$ don't decrease fast enough for its integral to have a finite value.

2. Evaluate $\int_{-\infty}^0 xe^x dx$.

Remark 2.3 (Notation). In computing improper integrals, it is tempting to write something like $f(\infty)$. The problem is, this expression makes no sense, because f is a function whose domain is some subset of $\mathbb{R}$, and ∞ is not a real number, so the function f has no value $f(\infty)$. That being said, I personally see little harm in *defining* the symbol $f(\infty)$ by $f(\infty) := \lim_{x\to\infty} f(x)$ as a shorthand. But, alas, math teachers everywhere seem to hate this, so unfortunately I have to ban you from writing things like $f(\pm\infty)$ and will most likely not give full credit if I see it on an exam :(

3. Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx$.

4. For what values of p is the integral

$$\int_1^\infty \frac{1}{x^p} dx$$

convergent?